ADVANCED CRYPTO TRACK

ACM India Summer School on Symmetric Cryptography · IIT Hyderabad

When Differentials Fail

Boomerang Cryptanalysis — in four acts

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June 15, 2026

Four rounds of AES.

No differential can break it.

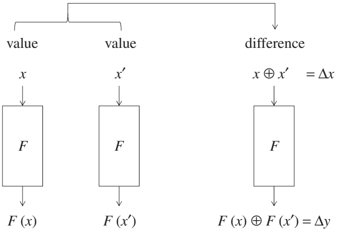
So the cipher is safe . . . right?

Act I · The Differential, Revisited

But first — what is a differential,

and why does it eventually fail?

The notion of difference



The idea

Take a *pair* of inputs with a fixed difference $\Delta = x \oplus x'$, push both through the cipher, and watch how that difference evolves.

One S-box is already random-ish

An input difference Δ_{in} does *not* fix the output difference.It induces a **distribution** over output differences:

$$\Pr[\Delta_{in} \xrightarrow{S} \Delta_{out}] = \frac{\#\{x : S(x) \oplus S(x \oplus \Delta_{in}) = \Delta_{out}\}}{2^n}$$

- Tabulating these counts gives the **Difference Distribution Table** (DDT).
- A good S-box keeps every DDT entry small (low differential uniformity).

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Stack the rounds: a characteristic

Fix the difference *after every round*. That chosen sequence is a **differential characteristic** (a trail):

$$\alpha = \Delta_0 \;\rightarrow\; \Delta_1 \;\rightarrow\; \Delta_2 \;\rightarrow\; \cdots \;\rightarrow\; \Delta_r = \beta$$

Assuming the rounds behave independently (the Markov-cipher assumption),

$$\Pr[\text{trail}] \;=\; \prod_{i=1}^{r} \Pr[\Delta_{i-1} \rightarrow \Delta_i].$$

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A differential is many trails

A **characteristic** pins every intermediate difference. A **differential** fixes *only* the two ends α and β.

Between the same α and β there are usually **many** characteristics — all the ways to get from α to β:

$$\Pr[\alpha \to \beta] = \sum_{\text{trails } \alpha \to \beta} \Pr[\text{trail}].$$

In practice

We cannot enumerate them all, so we approximate the differential by its **single best (dominant) trail** — a *lower bound* on the true probability.

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When is a differential actually useful?

A distinguisher must beat a coin flip. For a random permutation a fixed difference appears with probability $\approx 2^{-n}$.

The brute-force wall

A differential helps *only* while $\Pr[\alpha \to \beta] > 2^{-n}$.

- Every extra round multiplies in another factor < 1.
- Each active S-box costs up to a further $2^{-(\text{a few})}$.
- Stack enough rounds and the best trail **sinks below the wall**.

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Case study: four rounds of AES

- Block size $n = 128 \Rightarrow$ wall $= 2^{-128}$.
- Best single-S-box differential: 2^{-6}.
- Wide-trail theorem: any 4 rounds activate $\geq$ **25** S-boxes.

4-round bound
$\Pr[\text{trail}] \leq (2^{-6})^{25} = 2^{-150}$
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One trail cannot cross four rounds.

But what if we never asked it to span the whole cipher?

Act II · The Boomerang

If one trail can't reach,

can two short ones?

Wagner, 1999

Cut the cipher in half

Write the cipher as a composition of two halves:

$$E = E_1 \circ E_0$$

- Upper half E_0: a cheap differential $\alpha \to \beta$, prob p.
- Lower half E_1: a cheap differential $\gamma \to \delta$, prob q.

Each half spans *half* the rounds $\Rightarrow$ few active S-boxes $\Rightarrow$ p, q stay *far above* the wall.

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Why halves help

Suppose each half needs only $\sim$5 active S-boxes:

Whole cipher, one trail	Two half-trails
25 S-boxes $\Rightarrow 2^{-150}$	$p \approx q \approx 2^{-30}$
below the wall $\times$	each *far* above the wall $\checkmark$

The catch: a half-differential only covers *half* the cipher. We must **stitch** the two halves together — cleverly.

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The trick is in how we ask

An adaptive chosen-plaintext and chosen-ciphertext attack

We use the encryption *and* the decryption oracle — and *later* queries are built from *earlier* answers.

We work with a **quartet** of texts (P_1, P_2, P_3, P_4). The plan: force α on top, bounce δ at the bottom, and check whether α *comes back*.

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The boomerang algorithm, line by line

1. Pick P_1; set $P_2 = P_1 \oplus \alpha$.
2. **Encrypt**: $C_1 = E(P_1)$, $C_2 = E(P_2)$.
3. Set $C_3 = C_1 \oplus \delta, \quad C_4 = C_2 \oplus \delta$.
4. **Decrypt**: $P_3 = E^{-1}(C_3)$, $P_4 = E^{-1}(C_4)$.
5. **Test**: is $P_3 \oplus P_4 = \alpha$?

- ▷ a chosen-plaintext pair with difference α
- ▷ push the pair *forward* through E
- ▷ *adaptive* — shift both ciphertexts by δ
- ▷ a chosen-ciphertext query on the new pair
- ▷ if α returns far too often, we **distinguish**

Steps 3–4 are the heart: the ciphertext shift δ is the lower differential played *backwards*.

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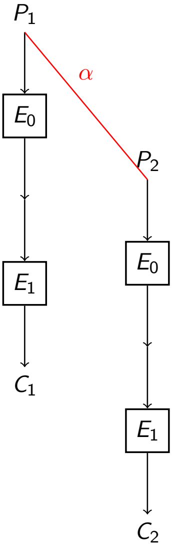
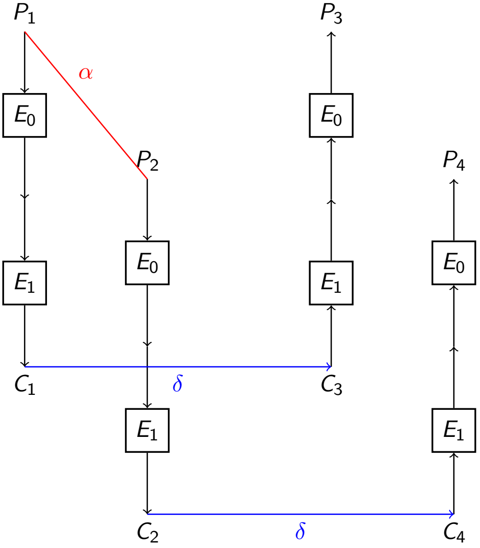
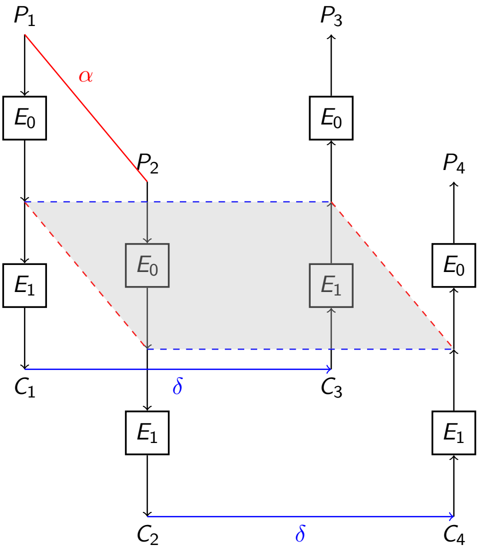


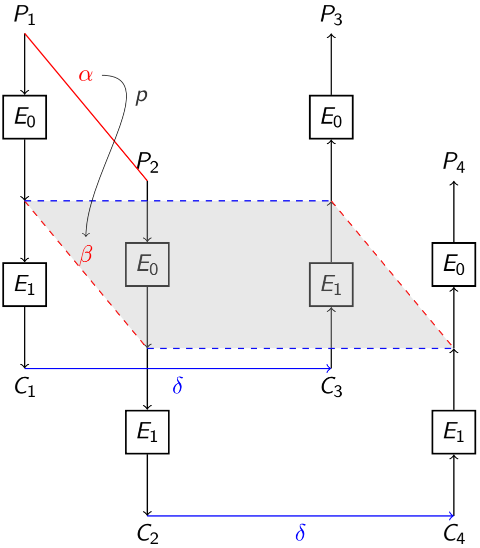
Note: $E = E_0 \circ E_1$



Same difference δ is added to the ciphertexts



The **Plane** of interest



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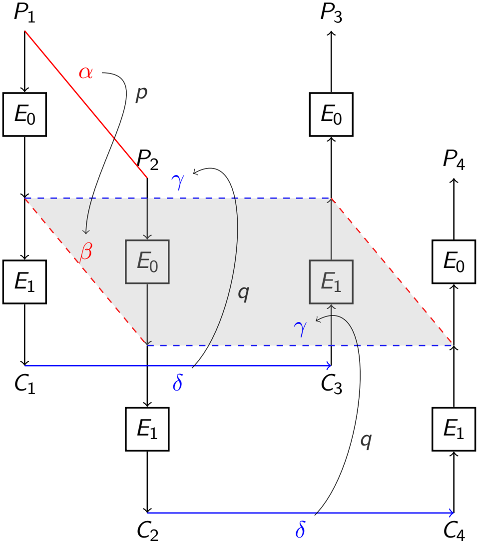
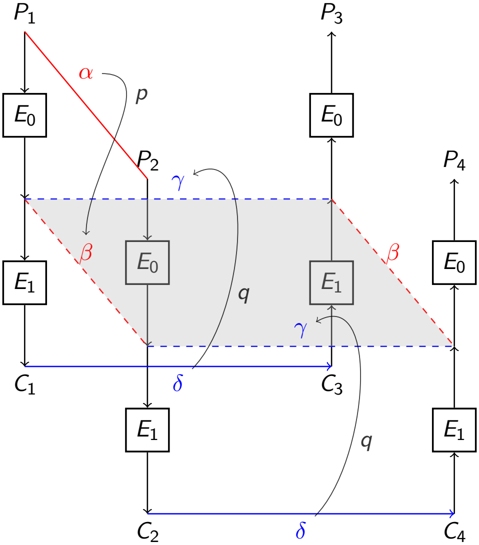
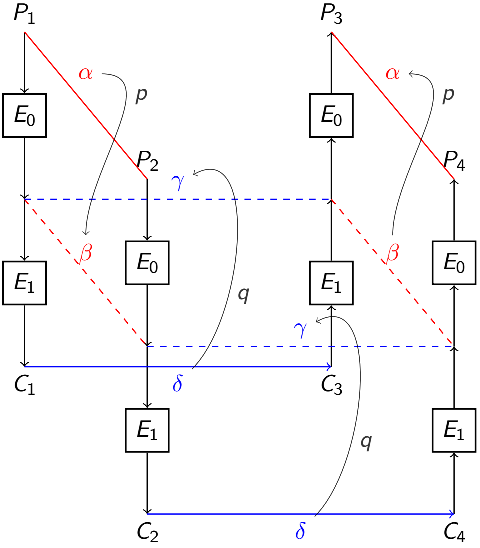


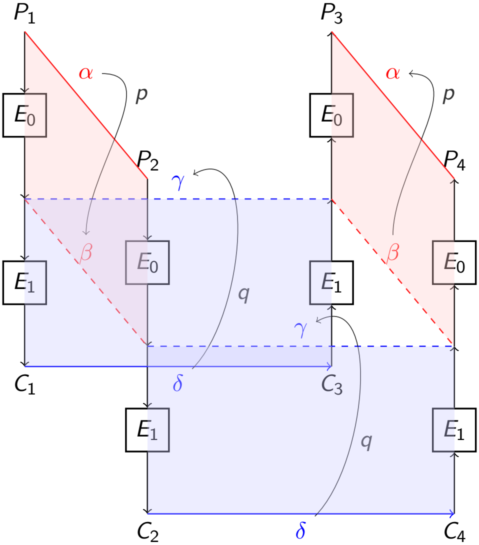
$$E_0 : P[\alpha \to \beta] = p \qquad E_1 : P[\gamma \to \delta] = q$$



γ on either side ensures β while decrypting



Hypothesis: $P_1 \oplus P_2 = P_3 \oplus P_4 = \alpha$



Probability of distinguisher: p^2q^2

Why the boomerang returns

Four differential events must line up in one quartet:

1. E_0 on (P_1, P_2): follows $\alpha \to \beta$ prob p
2. E_1 on (C_1, C_3): follows $\delta \to \gamma$ prob q
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Distinguisher probability

$$\Pr[\text{quartet}] = p \cdot q \cdot q \cdot p = p^2 q^2$$

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Above the wall again

One 4-round differential

$\leq 2^{-150} < 2^{-128}$

$\times$ hopeless

Boomerang: $2+2$ rounds

$p \approx q \approx 2^{-30}$

$p^2 q^2 \approx 2^{-120} > 2^{-128}$

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Splitting the cipher pushes the attack back above the wall.

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We just multiplied $p \cdot q \cdot q \cdot p$.

Were we allowed to?

Act III · The Seam Fights Back

The two halves meet in the middle.

What happens at the seam?

The fine print

We multiplied four probabilities into $p^2 q^2$. That step is valid *only* if the upper trail and the lower trail are **independent**.

The silent assumption

The β demanded from the top and the γ demanded from the bottom can be *freely combined* in the middle.

But the middle is a *deterministic* function. It does not negotiate.

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- Sean Murphy (2011): the independence assumption can fail *badly*.
- For a fixed quartet, the four texts share their middle values.
- The difference one trail needs may be **impossible** for the other — given those same values.

When the trails are incompatible

$$\Pr[\text{quartet}] = 0 \qquad (\text{not } p^2 q^2)$$

The estimate p^2q^2 was an **illusion** — the boomerang can *never* come back.

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The return of the cryptographic boomerang

- Sean Murphy (2011): the independence assumption can fail *badly*.
- For a fixed quartet, the four texts share their middle values.
- The difference one trail needs may be **impossible** for the other — given those same values.

When the trails are incompatible

$$\Pr[\text{quartet}] = 0 \qquad (\text{not } p^2q^2)$$

The estimate p^2q^2 was an **illusion** — the boomerang can *never* come back.

Where the contradiction lives

Walk one S-box at the seam for the whole quartet:

- the top trail fixes the pair of *values* entering the S-box,
- the bottom trail demands a particular difference *out* of it,
- but those values already *determine* the output difference.

Result

If the demanded and the determined differences disagree, *no* value of x can satisfy all four constraints — the quartet probability is exactly 0.

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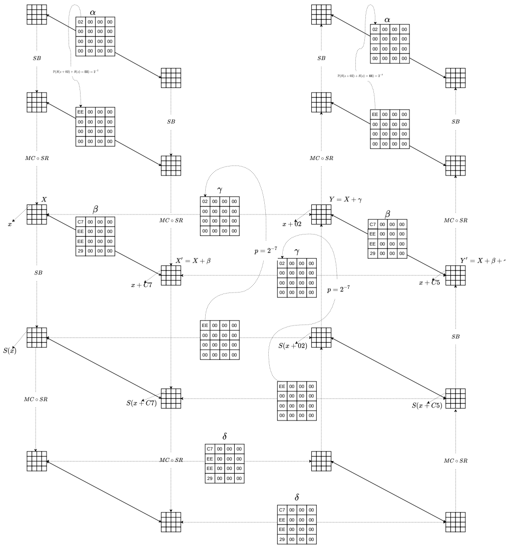
Murphy's incompatible trail for AES



Trace the quartet through the S-box layer. The byte differences forced from above and from below **contradict**.

No value of x satisfies all four constraints at once.

Result

Quartet probability $= \mathbf{0}$.

Your leaflet — and you can rotate it as a 3-D boomerang on the companion site.

A boomerang that never returns.

A bug in the attack . . .

or the seed of a sharper one?

Act IV · Turning Dependency Into a Weapon

What if the dependency at the seam

is not a bug, but a weapon?

Dependency cuts both ways

Murphy showed dependency can *kill* a boomerang ($\Pr = 0$).

The flip side

The same dependency can be **engineered in our favour**.

A family of follow-up tricks exploits it:

- the **S-box switch**
- the **ladder switch**
- the Feistel switch, the middle-round S-box trick . . .

all unified by the **sandwich** attack (Dunkelman–Keller–Shamir).

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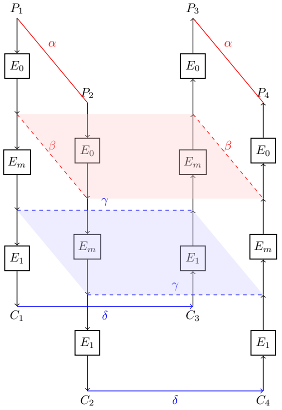
The sandwich: measure the seam

Stop pretending the seam is free.

Carve out a thin middle layer:

$$E = E_1 \circ E_m \circ E_0$$



- E_m is usually just *one round* (or one S-box layer).
- It carries the dependency, at a cost r.

Distinguisher

$$p^2 q^2 r$$

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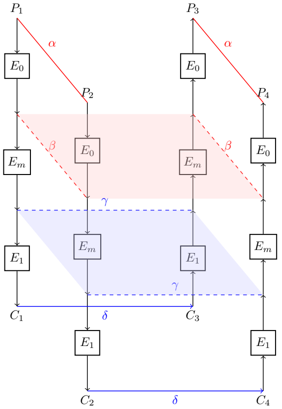

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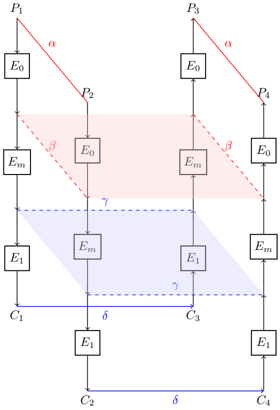

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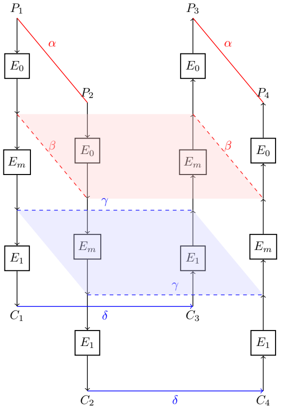

What exactly is r?

For one S-box at the seam, with input difference d and output difference c, r is the chance the boomerang *closes* across it:

$$r \;=\; \Pr_x\big[\, S^{-1}\big(S(x) \oplus c\big) \oplus S^{-1}\big(S(x \oplus d) \oplus c\big) = d \,\big]$$

This is one cell of a table

The **Boomerang Connectivity Table**: $\mathrm{BCT}(d, c) = 2^n \cdot r$.

Cid–Huang–Peyrin–Sasaki–Song, 2018 — one table that contains Murphy *and* both switches.

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Three kinds of cell tell three stories:

- $\mathrm{BCT} = 0$: the trails are **incompatible** — Murphy, $\Pr = 0$.
- $\mathrm{BCT} = 2^n$ (max): a **free** transition — the ladder switch, $r = 1$.
- in between: a paid but cheap transition — the S-box switch, $r = \mathrm{BCT}/2^n$.

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The whole story in one table

$d \backslash c$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
1	16	0	4	4	0	16	4	4	4	4	0	0	4	4	0	0
2	16	0	0	6	0	4	6	0	0	0	2	0	2	2	2	0
3	16	2	0	6	2	4	4	2	0	0	2	2	0	0	0	0
4	16	0	0	0	0	4	2	2	0	6	2	0	6	0	2	0
5	16	2	0	0	2	4	0	0	0	6	2	2	4	2	0	0
6	16	4	2	0	4	0	2	0	2	0	0	4	2	0	4	8
7	16	4	2	0	4	0	2	0	2	0	0	4	2	0	4	8
8	16	4	0	2	4	0	0	2	0	2	0	4	0	2	4	8
9	16	4	2	0	4	0	2	0	2	0	0	4	2	0	4	8
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C	16	0	6	0	0	4	0	6	2	2	2	0	0	0	2	0
D	16	2	4	2	2	4	0	6	0	0	2	2	0	0	0	0
E	16	0	2	2	0	0	2	2	2	2	0	0	2	2	0	0
F	16	8	0	0	8	0	0	0	0	0	0	8	0	0	8	16

BCT of the 4-bit PRESENT S-box (free value = 16)

- $d = 0$ or $c = 0$: free cross
- value 0: *incompatible* — Murphy
- in between: $r = \mathrm{BCT}/16$
- value 16: a *perfect* switch, $r = 1$

One picture — incompatibility, ladder switch and S-box switch, all at once.

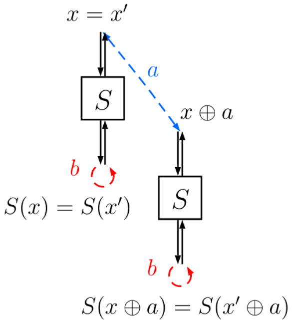
The ladder switch — transitions for free

Biryukov–Khovratovich: *route* the trails so a middle S-box sees a **vanishing** difference.

If its input $d_i = 0$ (or output $c_i = 0$):

$$\Pr\left[\,\cdots = d_i\,\right] = 1$$

That S-box costs nothing — it is the gold cross of the BCT.



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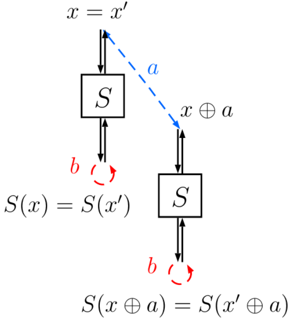

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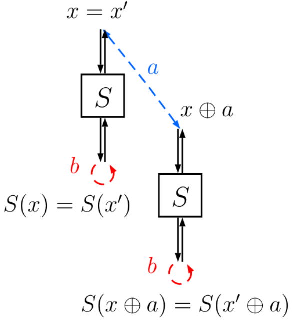

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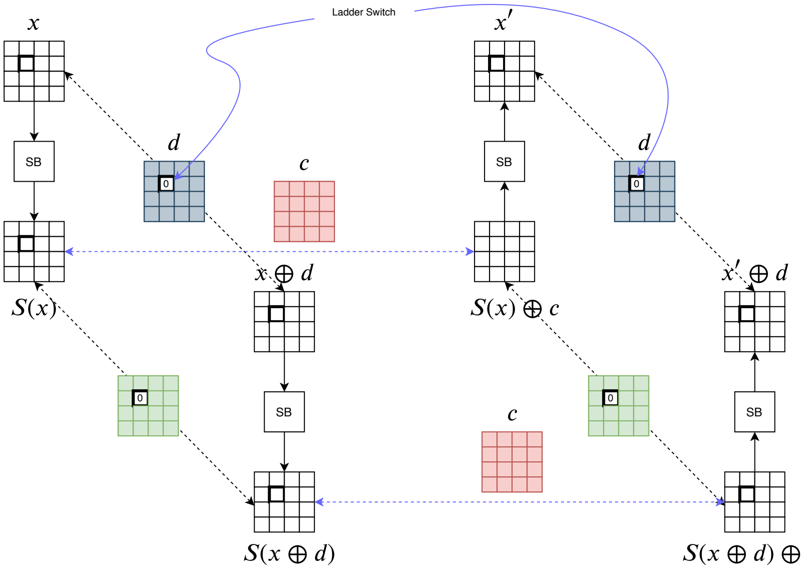
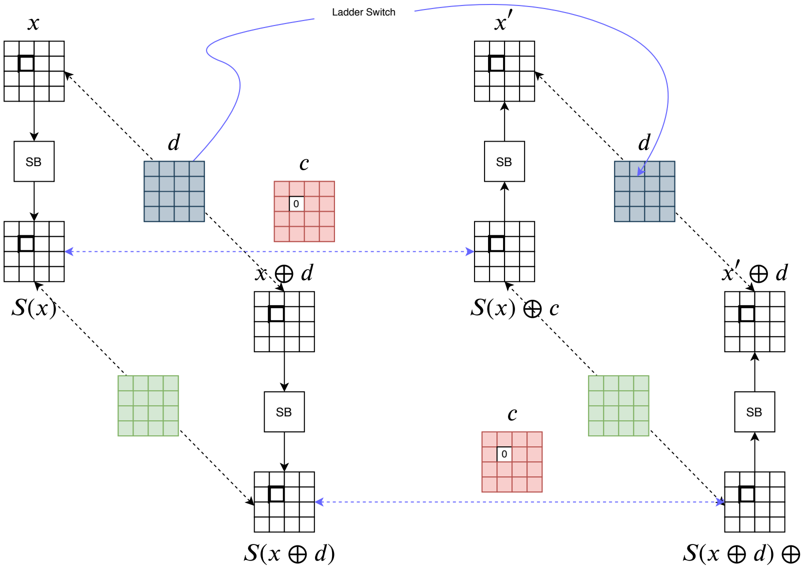

Ladder Switch
x
SB
S(x)
d
0
c
x ⊕ d
SB
S(x ⊕ d)
0
x′
SB
S(x) ⊕ c
d
0
x′ ⊕ d
SB
S(x ⊕ d) ⊕
0
c

Zooming - in ($\exists c_i = 0$)

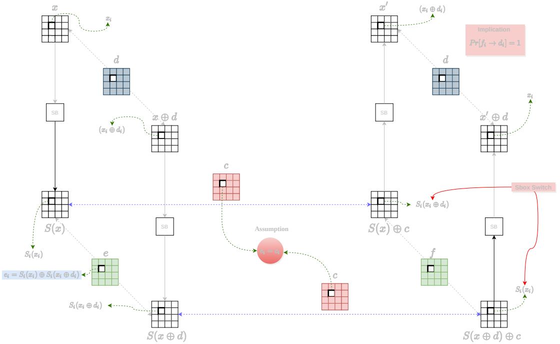


The S-box switch — cheap, not free

A genuine middle transition $d_i \to c_i$ with a *compatible* pair.

$$r_i = \frac{\mathrm{BCT}(d_i, c_i)}{2^n}$$

Far larger than the naive product would suggest — the dependency now **works for us**.



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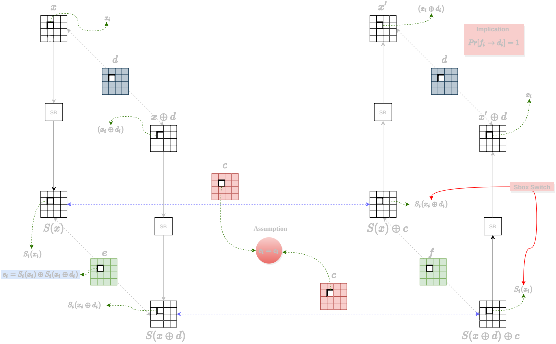

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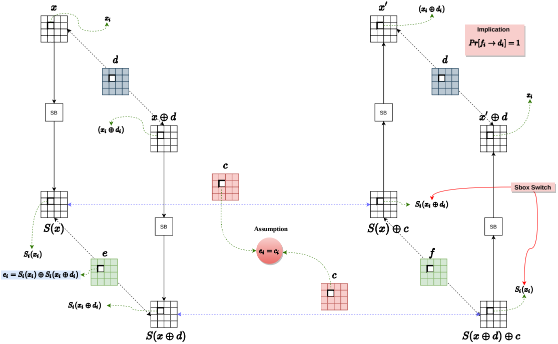

Four acts, one idea

The descent

1. Differentials sink below the wall.
2. Split the cipher $\Rightarrow$ boomerang p^2q^2.
3. The seam can make $\Pr = 0$ (Murphy).

The turn

4. Measure the seam: sandwich p^2q^2r.
5. The BCT: incompatibility, ladder & S-box switch, unified.

Take-away

Divide and conquer works on ciphers — *if you respect what happens at the seam.*

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Conclusion

- A cipher can be **provably safe against classical differentials** — and *still* fall.
- The **boomerang** buys this by splitting $E = E_1 \circ E_0$ and combining two short trails: $p^2 q^2$.
- That combination is **not free**: the seam can be **incompatible** (Murphy, $\Pr = 0$).
- The **sandwich** attack accounts for the seam exactly: $p^2 q^2 r$.
- The **Boomerang Connectivity Table** unifies it all — incompatibility, ladder switch, S-box switch — in a single object.

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Further Reading

- D. Wagner. *The Boomerang Attack*. FSE 1999.
- A. Biryukov, D. Khovratovich. *Related-key Cryptanalysis of the Full AES-192/256*. ASIACRYPT 2009. (ladder switch)
- O. Dunkelman, N. Keller, A. Shamir. *A Practical-Time Related-Key Attack on KASUMI*. CRYPTO 2010. (sandwich attack)
- S. Murphy. *The Return of the Cryptographic Boomerang*. IEEE Trans. IT, 2011. (incompatibility)
- C. Cid, T. Huang, T. Peyrin, Y. Sasaki, L. Song. *Boomerang Connectivity Table: A New Cryptanalysis Tool*. EUROCRYPT 2018. (BCT)

Going further — the Boomerang Difference Table (BDT)

S. Delaune, P. Derbez, M. Vavrille. *Catching the Fastest Boomerangs: Application to SKINNY*. ToSC 2020(2).

The BDT (and the Double-BDT) extend the BCT to track the dependency across *more than one* middle round, giving tighter boomerang probabilities. See also the *Feistel* counterpart, the FBCT (Boukerrou et al., ToSC 2020).

Essential Reading[1]: Symmetric Key Cryptography

1. **Serious Cryptography: A Practical Introduction to Modern Encryption**
   Jean-Philippe Aumasson
2. **The Block Cipher Companion**
   Lars R. Knudsen and Matthew J.B. Robshaw
3. **Security of Block Ciphers: From Algorithm Design to Hardware Implementation**
   Kazuo Sakiyama, Yu Sasaki, and Yang Li
4. **An Introduction to Mathematical Cryptography**
   Jeffrey Hoffstein, Jill Pipher, and J.H. Silverman

[1] In addition to other popular textbooks.

Explore every act live

Thank you

dhimans.tech/acmss-boomerang

Dr. Dhiman Saha · http://dhimans.in · dhiman@iitbhilai.ac.in

- de.ci.phe.red LAB
- Research group for focused work on crypto and security
- Spawned out of CS553: Cryptography course in 2018

http://de.ci.phe.red

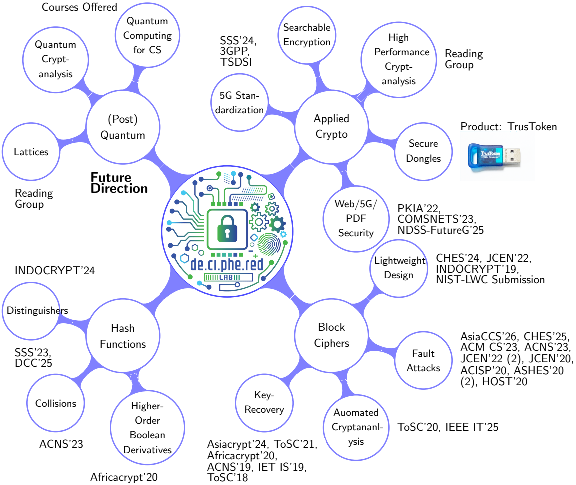
Current Composition

- 4 Post-docs, 7 PhD Students, 5 Masters, Multiple BTechs and Project Personnel

Research Overview

Collaborations

- ISI Kolkata
- CDAC Noida/Pune
- IIT Delhi
- IITKharagpur
- University of Essex
- University of New Brunswick
- NTU Singapore
- TU Graz
- Inria Paris
- IIITDM Kurnool
- L&T SCT
- LTI Mindtree



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